

Student Name:



2010

YEAR 11 PRELIMINARY YEARLY EXAMINATION

MATHEMATICS

EXTENSION 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 1 1/2 hours
- Write using blue or black pen
- Calculators may be used.
- Begin each question on a new page.
- All necessary working should be shown.

Total marks – 60

- Attempt Questions 1-5
- All questions are of equal value

Question 1 (12 Marks)		Use a Separate Sheet of paper	Marks
a)	In how many ways can the letters of the word BEGINNING be arranged in a line?		2
b)	A charm bracelet has 7 charms which are all different. How many distinct arrangements of the charms on the bracelet are possible?		2
c)	There are 16 students in a class, of whom 9 are male and 7 female. Five students are to be chosen for a working group to plan an excursion.		
	(i) In how many different ways can the group be chosen if there are no restrictions on its' membership?		1
	(ii) If the group must consist of 3 males and 2 females, in how many ways can the group be chosen?		1
d)	(i) Solve for x : $\frac{x+1}{x-3} \leq 3$		2
	(ii) Solve: $x^2 + 5x - 6 > 0$		2
	(iii) Hence, find values of x for which the following inequalities hold $\frac{x+1}{x-3} \leq 3$ and $x^2 + 5x - 6 > 0$		2

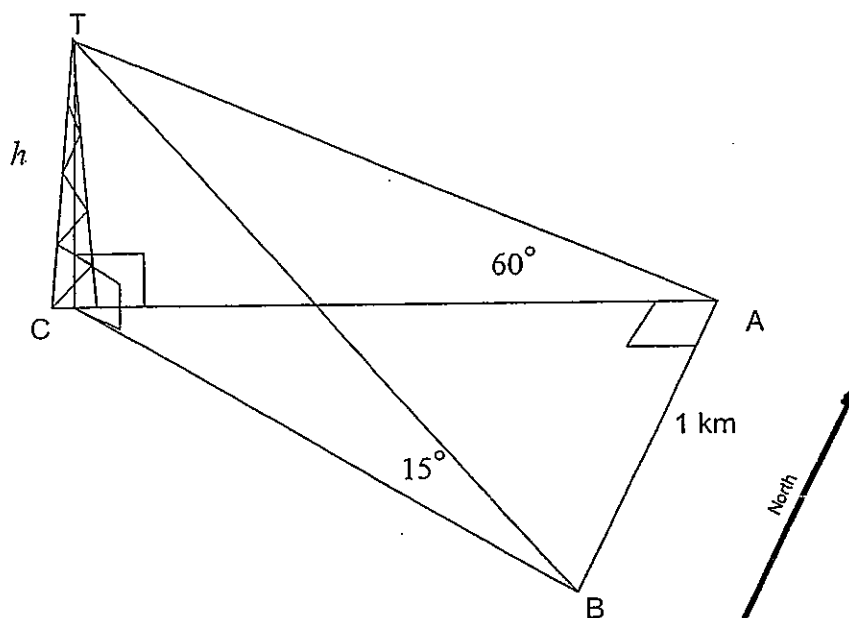
End of Question 1

Question 2 (12 Marks)

Use a Separate Sheet of paper

Marks

- a) (i) By expressing $\cos 3\theta$ as $\cos(\theta + 2\theta)$ show that
 $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2
- (ii) Hence solve the equation $\cos 3\theta = 2\cos^2 \theta - \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. 3
- b) (i) Express $4\sin \theta - 3\cos \theta$ in terms of $t = \tan \frac{\theta}{2}$. 2
- (ii) Hence solve $4\sin \theta - 3\cos \theta = 0$ for $-180^\circ \leq \theta \leq 180^\circ$. 2
- c) The angle of elevation of the top of a tower (T) from a point A due East of the tower is 60° . From a point B due South of A, the angle of elevation of T is 15° . A and B are at the same elevation as the base of the tower. If the distance $AB = 1$ km, find the height (h) of the tower. 3



End of Question 2

Question 3 (12 Marks)

Use a Separate Sheet of paper

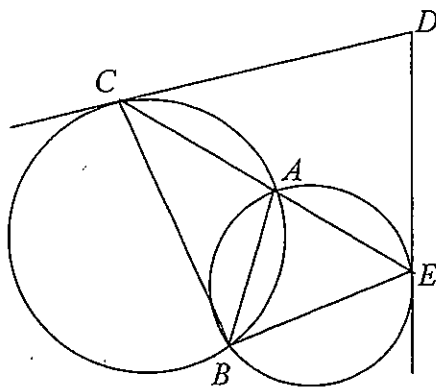
Marks

- a) Sketch the regions $y < \sqrt{9-x^2}$ and $y \geq x^2 - 9$ on the same graph, indicating where both regions hold simultaneously

2

b)

3



Two circles intersect at A and B . CAE is a straight line where C is a point on the first circle and E is a point on the second circle. The tangent at C to the first circle and the tangent at E to the second circle meet at D .

Prove that $BCDE$ is a cyclic quadrilateral.

- c) The polynomial $P(x) = x^3 + a^2x^2 + ax + b$ leaves a remainder of 2 when divided by x and a remainder of 13 when divided by $(x+1)$.

i) Show that $b = 2$

1

ii) Find the value of a .

2

- d) i) Express $3 \cos x + \sin x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 < \alpha < 90^\circ$ giving R in simplest form and α in degrees

2

ii) Hence solve the equation $3 \cos x + \sin x = -2$ for $0 \leq x \leq 360^\circ$ giving the solutions to the nearest degree.

2

End of Question 3

Question 4 (12 Marks)

Use a Separate Sheet of paper

Marks

- a) i) Expand $(a + \sqrt{b})^2$ 1
- ii) Given that $(a + \sqrt{b})^2 = 7 + 4\sqrt{3}$ find values for a and b . 2
- iii) Hence find $\sqrt{7 + 4\sqrt{3}}$ 1
- b) Calculate the acute angle between the lines
 $y = 2x - 3$ and
 $3x + 5y - 1 = 0$ to the nearest minute. 2
- c) Find the number of ways the letters of the word ANGLE can be arranged in a straight line so that
- i) No two consonants are next to each other 1
- ii) The three consonants are side-by-side 1
- iii) Exactly 2 of the 3 consonants are side-by-side 1
- d) Solve for x : $x^2 - 5x + 2 + \frac{4}{5x - x^2 - 2} = 0$ 3

End of Question 4

Question 5 (12 Marks)	Use a Separate Sheet of paper	Marks
a)	(i) Express $x^3 - x^2 - 10x - 8$ as a product of three linear factors.	2
	(ii) Solve the inequality $\frac{x^3 - 10x}{x^2 + 8} \geq 1$.	2
b)	$A(8, \sqrt{50})$ and $B(1, \sqrt{18})$ are two points. Find in simplest exact form the coordinates of the point P which divides the interval AB externally in the ratio $3 : 1$.	2
c)	The polynomial $P(x) = x^3 + 2x^2 - 4x - 1$ has zeros α , β and γ so that $P(x) = (x - \alpha)(x - \beta)(x - \gamma)$.	
	(i) Find the value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$.	1
	(ii) Find the value of $(\beta + \gamma)(\gamma + \alpha)(\alpha + \beta)$.	2
d)	A is the point $(5, 0)$ and O is the origin. Given that the point B (x, y) lies on the line $y = 1 - 3x$ and that OB is perpendicular to AB, find all possible coordinates of B.	3

End of Examination

2010 Year 11 Ext 1 Preliminary HSC

Question 1

a) $\frac{9!}{3!2!2!} =$

b) $\frac{6!}{2}$

c) i) ${}^{16}C_5 =$

ii) ${}^9C_3 \times {}^7C_2 =$

d) i) $\frac{x+1}{x-3} \cdot x(x-3)^2 \leq 3(x-3)^2$

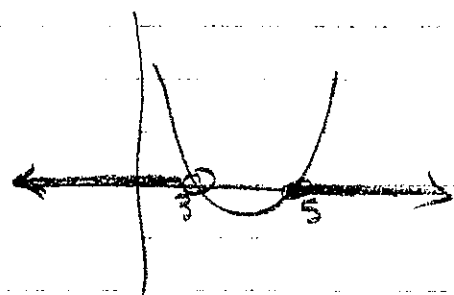
$$(x+1)(x-3) \leq 3(x-3)^2$$

$$0 \leq 3(x-3)^2 - (x+1)(x-3)$$

$$0 \leq (x-3) \{ 3x-9-x-1 \}$$

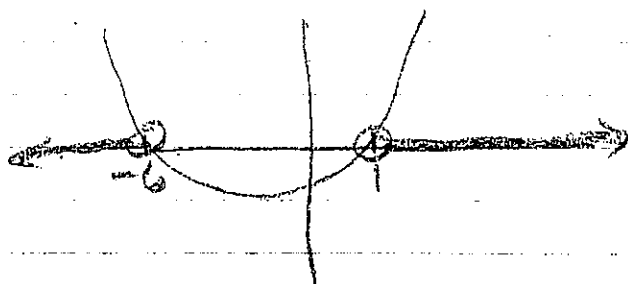
$$0 \leq (x-3)(2x-10)$$

$$0 \leq 2(x-3)(x-5)$$

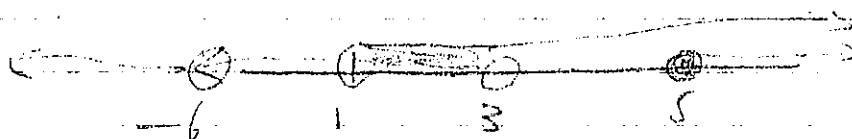


∴ $x < 3$ or $x \geq 5$

i) $(x+6)(x-1) > 0$



iii) $x \geq 5$ or $x < -6$ or $1 < x < 3$



Question 2

a) $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
 $\cos(\theta + 2\theta) = \cos \theta \cos 2\theta - \sin \theta \sin 2\theta$
 $= \cos \theta (2\cos^2 \theta - 1) - \sin \theta (2\sin \theta \cos \theta)$
 $= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta$
 $= 2\cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta) \cos \theta$
 $= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta$
 $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$

ii) $\cos 3\theta = 2\cos^2 \theta - \cos \theta$
 $4\cos^3 \theta - 3\cos \theta = 2\cos^2 \theta - \cos \theta$
 $4\cos^3 \theta - 2\cos^2 \theta - 2\cos \theta = 0$
 $2\cos \theta (2\cos^2 \theta - \cos \theta - 1) = 0$

$2\cos \theta (2\cos^2 \theta + 1)(\cos \theta - 1) = 0$
 $\cos \theta = 0$ or $\cos \theta = -\frac{1}{2}$ or $\cos \theta = 1$

~~$2\cos \theta + 1$~~
 ~~$\cos \theta - 1$~~

$\theta = 90^\circ, 270^\circ, \theta = 120^\circ, 240^\circ, \theta = 0^\circ \text{ or } 360^\circ$

b) $\sin \theta = \frac{2t}{1+t^2}, \cos \theta = \frac{1-t^2}{1+t^2}$

$4\sin \theta - 3\cos \theta = \frac{8t}{1+t^2} - \frac{3(1-t^2)}{1+t^2}$

$4\sin \theta - 3\cos \theta = 0$

Check $\theta = 180^\circ$?
 $4 \times 0 - 3 \times 1 \neq 0$

$\frac{8t}{1+t^2} - \frac{3(1-t^2)}{1+t^2} = 0$

$8t - 3 + 3t^2 = 0$

$3t^2 + 8t - 3 = 0$

$(3t-1)(t+3) = 0$

$t = \frac{1}{3}$ or -3

~~$3t - 1$~~
 ~~$t + 3$~~

$$\tan \frac{\theta}{2} = \frac{1}{3}$$

or

$$\tan \frac{\theta}{2} = -3$$

$$\frac{\theta}{2} = 18.43^\circ$$

$$\theta = 36^\circ 52'$$

$$\frac{\theta}{2} = -71.565^\circ$$

$$\theta = -143^\circ 8'$$

$$c) \quad \frac{AC}{h} = \tan 30^\circ \Rightarrow AC = h \tan 30^\circ$$

$$\frac{BC}{h} = \tan 75^\circ \Rightarrow BC = h \tan 75^\circ$$

$$BC^2 = AC^2 + AD^2$$

$$h^2 \tan^2 75^\circ = h^2 \tan^2 30^\circ + 1$$

$$h^2 (\tan^2 75^\circ - \tan^2 30^\circ) = 1$$

$$h^2 =$$

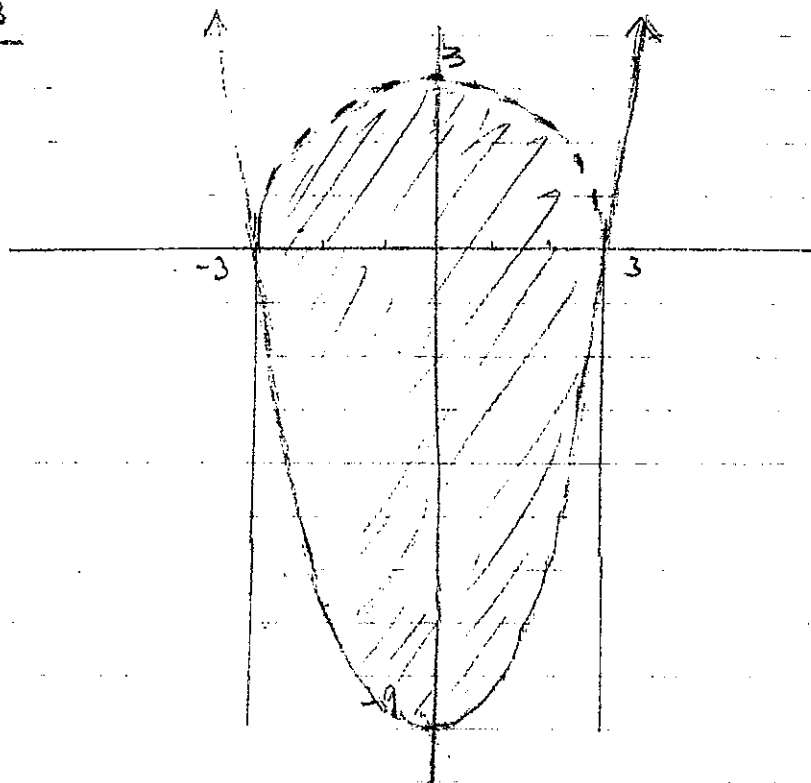
$$\frac{1}{\tan^2 75^\circ - \tan^2 30^\circ}$$

$$h = \sqrt{0.073557} \text{ km}$$

$$h = 73.557 \text{ m}$$

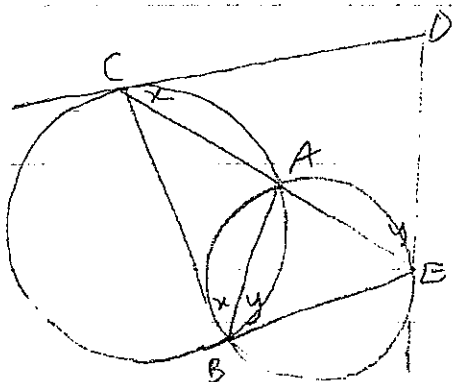
Question 3

a)



2

b)



$\angle DCE = \angle CBA = x$
(angle between tangent & chord equals angle in alternate segment)

$\angle DEC = \angle ABE = y$
(alternate segment theorem)

$\angle CDE = 180 - (x+y)$ (angle sum $\nabla \triangle CDE$)

Also $\angle CBE = x+y$

$\therefore \angle CDE + \angle CBE = 180 - (x+y) + x+y$
 $= 180^\circ$

$\therefore BCDE$ is a cyclic quadrilateral
(one pair of opposite angles are supplementary)

3

$$c) \quad P(x) = x^3 + a^2 x^2 + ax + b$$

$$P(0) = 2$$

$$2 = 0 + 0 + 0 + b$$

$$\therefore P(x) = x^3 + a^2 x^2 + ax + 2$$

$$P(-1) = 13$$

$$13 = -1 + a^2 - a + 2$$

$$0 = a^2 - a - 12$$

$$0 = (a-4)(a+3)$$

$$a = 4 \text{ or } -3$$

$$P(x) = x^3 + 16x^2 + 4x + 2$$

$$\text{check } P(-1) = 13$$

$$\sim P(x) = x^3 + 9x^2 - 3x + 2$$

$$P(-1) = 13$$

$$d) \quad i) \quad 3 \cos x + \sin x = R \sin(x + \alpha)$$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$\therefore R \cos \alpha = 1 \quad \text{--- (1)}$$

$$R \sin \alpha = 3 \quad \text{--- (2)}$$

$$\textcircled{1}^2 + \textcircled{2}^2$$

$$R^2 = 3^2 + 1^2$$

$$R = \sqrt{10}$$

$$\tan \alpha = 3 \quad \Rightarrow \quad \alpha = \tan^{-1}(3) \doteq 71^\circ 34'$$

$$\sqrt{10} \sin(x + 71^\circ 34') = -2$$

$$\sin(x + 71^\circ 34') = -\frac{2}{\sqrt{10}}$$

$$x + 71^\circ 34' = 180 + 39^\circ 14' \text{ or } 360 - 39^\circ 14'$$

$$x = 147^\circ 40' \text{ or } 249^\circ 12'$$

$$x \doteq 148^\circ \text{ or } 249^\circ$$

to nearest degree

$$\frac{4}{a)} i) (a + \sqrt{b})^2 = a^2 + 2a\sqrt{b} + b$$

$$ii) 7 + 4\sqrt{3} = a^2 + b + 2a\sqrt{b}$$

$$\therefore a^2 + b = 7$$

$$2a\sqrt{b} = 4\sqrt{3} \Rightarrow 4a^2b = 48$$

$$a^2b = 12$$

$$a^2 = 7 - b$$

$$(7 - b)b = 12 \Rightarrow 7b - b^2 = 12$$

$$b^2 - 7b + 12 = 0$$

$$(b - 3)(b - 4) = 0$$

$$b = 3 \text{ or } 4$$

$$\therefore 3a^2 = 12$$

$$a^2 = 4$$

$$a = \pm 2$$

$$\text{or } 4a^2 = 12$$

$$a^2 = 3$$

$$a = \pm\sqrt{3}$$

$$\therefore \underline{a = \pm 2 \quad b = 3}$$

$$\text{or } \underline{a = \pm\sqrt{3} \quad b = 4}$$

iii)

$$\sqrt{7 + 4\sqrt{3}} = 2 + \sqrt{3} \quad \text{for positive square root}$$

$$\text{or } \sqrt{7 + 4\sqrt{3}} = \sqrt{3} + \sqrt{4} \quad \text{gives same answer}$$

b)

$$y = 2x - 3$$

$$m_1 = 2$$

$$3x + 5y - 1 = 0$$

$$m_2 = -\frac{3}{5}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2 + \frac{3}{5}}{1 - 2 \times \frac{3}{5}} \right| = \left| \frac{13}{-1} \right|$$

$$\tan \theta = 13$$

$$\theta \doteq 85^\circ 36'$$

c) ANGLE
 i) No two consonants CVCVC
 $3! \times 2! =$

ii) CCCVV
 group consonants $3!$ ways
 then arrange CCCVV $3!$ ways
 $\therefore 3! \times 3!$

iii) Exactly 2 is complement of (i) + (ii)
 $\therefore$ Total arrangements = $5!$

$$\begin{aligned} \text{Exactly 2 consonants} &= 5! - 3! \times 2! - 3! \times 3! \\ &= 120 - 12 - 36 \\ &= 72 \text{ ways} \end{aligned}$$

d) $x^2 - 5x + 2 = \frac{4}{x^2 - 5x + 2} = 0$

Put $u = x^2 - 5x + 2$

$$u - \frac{4}{u} = 0$$

$$u^2 - 4 = 0$$

$$u = \pm 2$$

$$\therefore x^2 - 5x + 2 = 2$$

$$\checkmark x^2 - 5x + 2 = -2$$

$$x(x-5) = 0$$

$$x = 0 \text{ or } 5$$

$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

$$x = 4 \text{ or } 1$$

$$x = 0, 5, 4, 1$$

Question 5

a) i) $P(x) = x^3 - x^2 - 10x - 8$

$$P(1) = 1 - 1 - 10 - 8 \neq 0$$

$$P(-1) = -1 - 1 + 10 - 8 = 0$$

$$\begin{array}{r} x^2 - 2x - 8 \\ x+1 \overline{) x^3 - x^2 - 10x - 8} \\ \underline{x^3 + x^2} \\ -2x^2 - 10x - 8 \\ \underline{-2x^2 - 2x} \\ -8x - 8 \\ \underline{-8x - 8} \\ 0 \end{array}$$

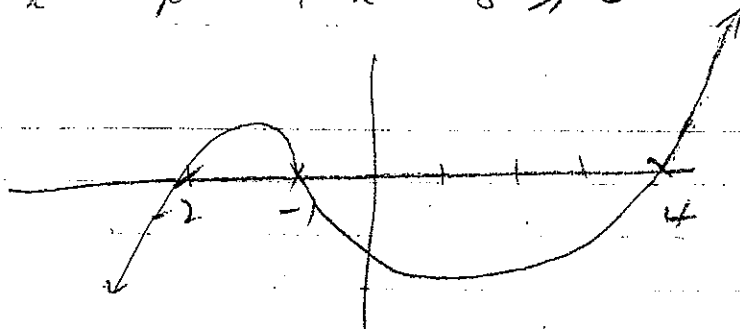
$$P(x) = (x+1)(x^2 - 2x - 8) = (x+1)(x-4)(x+2)$$

ii) $\frac{x^3 - 10x}{x^2 + 8} \geq 1$

Note $x^2 + 8$ is always +

$$x^3 - 10x \geq x^2 + 8$$

$$x^3 - x^2 - 10x - 8 \geq 0$$



$$-2 \leq x \leq -1 \quad \text{or} \quad x \geq 4$$

b) A $(x_1, y_1) = (8, \frac{5}{2})$ B $(x_2, y_2) = (1, \frac{3}{2})$

$$\frac{1}{2} = \frac{3}{-1}$$

$$x = \frac{3 \times 1 + -1 \times 8}{3 - 1}$$

$$y = \frac{3 \times \frac{3}{2} + -1 \times \frac{5}{2}}{3 - 1}$$

$$x = -\frac{5}{2}$$

$$= \frac{4\sqrt{2}}{2} = 2\sqrt{2}$$

$$P \left(-\frac{5}{2}, 2\sqrt{2} \right)$$

c)

$$p(x) = x^3 + 2x^2 - 4x - 1$$

$$= (x-\alpha)(x-\beta)(x-\gamma)$$

i)

For $(1-\alpha)(1-\beta)(1-\gamma)$ Sub $x=1$

$$p(1) = 1 + 2 - 4 - 1 = -2$$

ii)

For $(\beta+\gamma)(\gamma+\alpha)(\alpha+\beta)$

$$p(x) = (x-\alpha)(x-\beta)(x-\gamma)$$

Sub $x = \alpha + \beta + \gamma$ $p(\alpha + \beta + \gamma) = (\beta + \gamma)(\gamma + \alpha)(\alpha + \beta)$

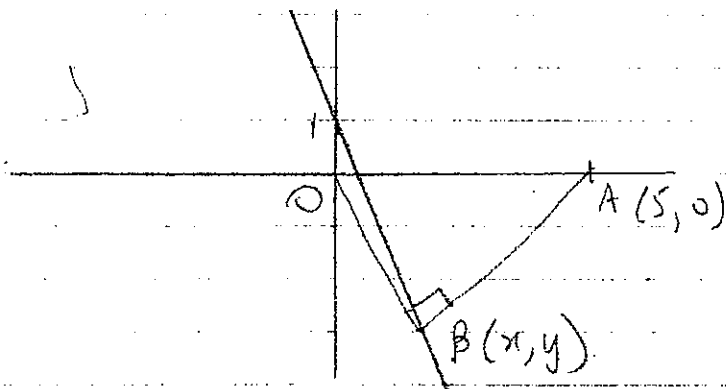
$$= (\alpha + \beta + \gamma)^3 + 2(\alpha + \beta + \gamma)^2 - 4(\alpha + \beta + \gamma) - 1$$

but $(\alpha + \beta + \gamma) = -2$

$$p(\alpha + \beta + \gamma) = (-2)^3 + 2(-2)^2 - 4(-2) - 1 = -8 + 8 + 8 - 1$$

$$= \underline{\underline{7}}$$

d)



For OB, $m_1 = \frac{y}{x}$

For AB, $m_2 = \frac{y-0}{x-5}$

$$m_1 m_2 = -1$$

$$\frac{y}{x} \cdot \frac{y}{x-5} = -1$$

$$y^2 = -x(x-5)$$

$$= -x^2 + 5x$$

$$x^2 - 5x + y^2 = 0$$

$$y = 1 - 3x$$

$$x^2 - 5x + (1 - 3x)^2 = 0$$

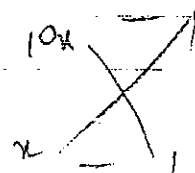
$$x^2 - 5x + 1 - 6x + 9x^2 = 0$$

$$10x^2 - 11x + 1 = 0$$

$$(10x-1)(x-1) = 0$$

$$x = \frac{1}{10} \text{ or } 1$$

$$y = \frac{7}{10} \text{ or } -2$$



B is $(\frac{1}{10}, \frac{7}{10})$
or $(1, -2)$